Data Communication and Computer Networks

Chapter 2 | Modulation, Multiplexing, Media and Switching

Unit 2 - Analog transmission, multiplexing, transmission media, circuit and packet switching (8 hrs)

Syllabus covered

Analog transmission - modulation of digital data (ASK, FSK, PSK, QAM) and modulation of analog signals (AM, FM, PM); multiplexing - FDM, WDM, TDM; transmission media - guided (twisted pair, coaxial, fiber) and unguided (radio, microwave, infrared); circuit switching and packet switching.

What is inside

Section	Question type	Marks	Count
Section A	Short answer - definitions, one-liners, small numericals	2	42
Section B	Long answer - full explanations with tables and diagrams	7	10

How to use this bank

- Learn the three carrier parameters - amplitude, frequency, phase - because every digital-to-analog scheme varies exactly one of them.
- For each scheme remember one number: ASK and PSK need B = (1+d)S, FSK needs more, AM needs 2B.
- In multiplexing questions the guard band is always the trap: n channels need n-1 guard bands.
- Draw every switching answer as three phases - setup, data transfer, teardown.

Source tags

- **PYQ <year>** - asked in a C. V. Raman Global University end-semester paper (2022, 2023, 2024-25 or the 2024-25 backlog paper).
- **Model** - from the printed unit-wise model question paper.
- **Assignment** - from the course assignment for this unit.
- **Notes** - from the lecture slides, not yet asked but examinable.

Section A | 2-mark questions

Answer in two to four lines. Each point below is one mark-bearing idea, so a single line written in the exam earns the mark.

A1. Illustrate the requirement of analog-to-analog modulation techniques. [Model]

- Analog data can be sent baseband only over a **low-pass channel**.
- Most long-distance media - radio, microwave, satellite, the telephone local loop - offer only a **bandpass channel**, so the signal must be shifted to a carrier frequency.
- It also lets many signals share one medium on different carriers, which is FDM.

A2. Discuss the requirement of analog-to-digital conversion approaches. [Model]

- Digital signals are **more robust to noise**, and can be regenerated, corrected, compressed, encrypted and stored.
- Analog sources such as voice and video must therefore be digitised by **PCM or delta modulation** before a digital link can carry them.
- Digitising also allows TDM, error control and integration with computer data.

A3. Explain the need for data encoding before sending over a medium. [Model]

- A medium carries **electromagnetic signals**, not bits; encoding maps bits to signal levels or onto a carrier.
- **Line coding** produces a baseband signal suited to the medium and provides synchronisation.
- **Modulation** shifts the data into a band the medium can pass.
- Good encoding also removes the DC component and packs more bits per symbol.

A4. What are the possible analog-to-analog modulation techniques? [Model]

- **AM** - amplitude of the carrier varies.
- **FM** - frequency of the carrier varies.
- **PM** - phase of the carrier varies.

A5. Compare the bandwidth requirement of the three analog-to-analog modulation techniques. [Model]

Scheme	Bandwidth	Remark
AM	2B	Carrier plus two sidebands; the cheapest in bandwidth
FM	2(1 + beta)B	(1 + beta) times that of AM; beta = modulation index
PM	2(1 + beta)B	Same width as FM; differs only in how the index is defined

A6. Between AM and FM, which one gives better noise immunity? [Model]

- **FM** is far more immune.
- Noise adds to the **amplitude**, and AM carries its information in the amplitude, so the noise goes straight into the data.
- FM carries information in **frequency**, so a limiter can clip the amplitude noise away without touching the data.
- Price: FM needs (1 + beta) times the bandwidth.

A7. What are the possible digital-to-analog modulation techniques? [Model]

- **ASK** - amplitude of the carrier is varied.
- **FSK** - frequency of the carrier is varied.
- **PSK** - phase of the carrier is varied.
- **QAM** - amplitude and phase varied together, so more bits per symbol.

A8. Why is PSK preferred as the modulation technique in modems? [Model]

- Noise changes **amplitude** far more easily than it changes phase, so PSK has the best noise immunity of the three.
- It does not need **two carrier signals**, unlike FSK.
- It uses the **same bandwidth as ASK** - (1 + d)S - while being more robust.
- Multilevel PSK (QPSK, 8-PSK) carries 2 or 3 bits per symbol, which is why voice-band modems used it.

A9. Out of the three digital-to-analog techniques, which one requires higher bandwidth? [Model]

- **FSK** requires the highest bandwidth.
- It needs **two carrier frequencies**, so the band must be at least 2 x delta-f plus (1 + d)S.
- ASK and PSK need only (1 + d)S, where d depends on the modulation and filtering.

A10. Define carrier signal and state its role in analog transmission of digital data. [PYQ 2022; Model]

- A carrier is a **high-frequency analog signal** of frequency fc whose amplitude, frequency or phase is varied by the data.
- It **transports** the digital information through a bandpass medium - nothing is transmitted without it.
- It allows **frequency translation**, so many users can share a medium on different carriers.
- One sentence for the exam: the data is the passenger, the carrier is the vehicle.

A11. Describe frequency modulation with a waveform. [PYQ 2024-25 backlog]

- The **frequency** of the carrier changes with the amplitude of the modulating signal; the amplitude of the carrier stays constant.
- A large modulating amplitude gives a large frequency deviation, so the cycles become **dense**; a small amplitude gives sparse cycles.
- Bandwidth = 2(1 + beta)B; it is the standard for FM broadcast, where each station gets 200 kHz.

A12. Define analog-to-analog conversion. [PYQ 2024-25 backlog]

- It is the representation of analog information, such as voice, by an analog signal - the amplitude, frequency or phase of a carrier is changed to follow the information.
- It is needed when the available channel is **bandpass** rather than low-pass.

A13. Define constellation diagram and explain its role in analog transmission. [PYQ 2024-25 backlog]

- A two-dimensional plot of the signal points of a modulation scheme: the x-axis is the **in-phase** (I) component, the y-axis the **quadrature** (Q) component.
- Distance from the origin gives the **amplitude**; the angle gives the **phase**.
- With L points, each symbol carries log2L bits.
- The **minimum distance between points** measures noise immunity - closer points are easier to confuse.

A14. Which of AM, FM and PM is most susceptible to noise? Defend your answer. [PYQ 2024-25 backlog]

- **AM** is the most susceptible.
- Noise is an unwanted addition to the signal, and it adds to the **amplitude**, which is exactly the parameter AM uses to carry information.
- FM and PM carry information in frequency and phase and have a constant envelope, so amplitude noise can be clipped away.

A15. Calculate the bandwidth needed to modulate a 5 kHz voice signal using AM, and state the number of channels allocated by the FCC. [PYQ 2024-25]

1. BAM = 2 x B = 2 x 5 kHz = **10 kHz** per station.
2. The AM band is 535 to 1605 kHz, that is 1070 kHz, so at most 107 channels of 10 kHz fit.
3. The **FCC has allocated 100 channels** for AM broadcast, leaving the rest as spacing.

A16. Explain the need for modulation of digital data. [PYQ 2024-25]

- Digital data are baseband; a **bandpass medium** - telephone line, radio, satellite - cannot carry them directly.
- Modulation shifts the data onto a carrier inside the passband of the medium.
- It also makes **antennas practical**: a baseband signal would need an antenna kilometres long, whereas a carrier of MHz range needs a fraction of a metre.
- It allows **FDM**, so many users share one medium.

A17. Define bit rate and baud rate in analog transmission, and give the relation between them. [Notes]

- **Bit rate N** - number of bits per second, in bps.
- **Baud rate (signal rate) S** - number of signal elements per second, in baud.
- **Relation**: S = N x 1/r, where r is the number of data bits carried by one signal element.
- In analog transmission the baud rate is **less than or equal to** the bit rate.

A18. An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate. [Notes]

1. r = 4, S = 1000 baud.
2. N = S x r = 1000 x 4 = **4000 bps = 4 kbps**.

A19. An available bandwidth of 100 kHz spans 200 to 300 kHz. Find the carrier frequency and the bit rate for ASK with d = 1. [Notes]

1. The carrier sits in the middle of the band: **fc = 250 kHz**.
2. B = (1 + d)S, so 100 kHz = 2S and S = 50 kbaud.
3. With r = 1, N = S x r = **50 kbps**.

A20. What is BASK and how is it generated? [Notes]

- In binary ASK the **amplitude** of the carrier is switched: bit 1 sends the carrier, bit 0 sends zero voltage.
- Generated by **multiplying the unipolar NRZ signal** with the carrier.
- Bandwidth B = (1 + d)S; simple, but weak against noise.

A21. What is BFSK and how is it implemented? [Notes]

- In binary FSK the **frequency** is switched: f1 = fc + delta-f for 1 and f2 = fc - delta-f for 0.
- Implemented by feeding **NRZ into a VCO** (voltage-controlled oscillator), whose output frequency follows the input voltage.
- Amplitude is constant, so it is immune to amplitude noise, but it needs the widest band.

A22. Explain binary phase shift keying. [PYQ 2024-25 backlog]

- In BPSK the **phase** of the carrier takes one of two values, 0 and 180 degrees: 1 sends phase 0, 0 sends phase 180.
- Generated by multiplying the **polar NRZ** signal with the carrier - a negative NRZ level simply inverts the carrier.
- Bandwidth is the same as ASK, (1 + d)S, but noise immunity is the best of the three, because the two points are 180 degrees apart on the constellation.

A23. Why is PSK superior to FSK? [Notes]

- It does not need **two carrier frequencies**, so it uses less bandwidth.
- Phase is less affected by noise than amplitude, so error performance is better.
- It scales to multilevel PSK and QAM, carrying several bits per symbol.
- The cost is **more complex hardware** to detect phase.

A24. What is QAM? [Notes]

- Quadrature amplitude modulation varies **both amplitude and phase** of the carrier, combining ASK and PSK.
- With L amplitude levels and M phases it gives L x M points, so log2(L x M) bits per symbol.
- Bandwidth is the same as ASK, which is why 16-QAM and 64-QAM are used in modems and digital TV.

A25. How do guided media differ from unguided media? [Model]

Point	Guided	Unguided
Conduit	Physical cable guides the signal	No cable - free space
Examples	Twisted pair, coaxial, fiber	Radio, microwave, infrared
Direction	Point-to-point along the cable	Broadcast or beam through the air
Security	Confined, harder to tap	Radiates, anyone in range can receive
Impairment	Attenuation, crosstalk	Attenuation, interference, multipath

A26. Write the significance of twisting in a twisted-pair cable. [Model]

- Two insulated copper wires are twisted into a helix, one carrying the positive and one the negative signal.
- The twists make the **induced noise on the two wires equal and opposite**, so it cancels in the differential receiver.
- Twisting also reduces **crosstalk** between neighbouring pairs.
- More twists per metre means better quality - the basis of Category 5e and Category 6 cable.

A27. What are the three major classes of guided media? [Model]

- **Twisted-pair cable** - UTP and STP; telephone and Ethernet.
- **Coaxial cable** - inner conductor, dielectric, braided shield, outer cover; cable TV and broadband.
- **Fiber-optic cable** - glass or plastic core carrying light by total internal reflection; backbones and long haul.

A28. Mention the advantages of optical fiber over twisted pair and coaxial cable. [PYQ 2022; PYQ 2023; PYQ 2024-25; Model]

- **Very high bandwidth** - terabits per second, far beyond copper.
- **Low attenuation**, so repeaters are needed only every tens of kilometres.
- **Immune to electromagnetic interference and crosstalk**, because the signal is light, not current.
- **No spark hazard** and no corrosion; lighter and thinner than copper.
- **More secure** - hard to tap without breaking the fiber.

A29. Explain the purpose of cladding in an optical fiber. [Model]

- The core is surrounded by cladding of **lower refractive index**.
- Light travelling in the core that strikes the boundary at more than the critical angle is **totally internally reflected** back into the core, so it follows the fiber instead of escaping.
- The cladding also **protects the core** and keeps the light path clean.

A30. Which medium generally offers better security: guided or unguided? Justify your answer. [PYQ 2024-25]

- **Guided** media are more secure.
- The signal is **confined inside the cable**, so an eavesdropper needs physical access to the cable, and tapping disturbs the signal and is detectable.
- Unguided media **radiate into space**: any receiver in range with a suitable antenna can pick up the signal.
- Within guided media, optical fiber is the most secure of all.

A31. In what situation is multiplexing used? [Model]

- Whenever the **bandwidth of a link is greater than the bandwidth needed by one signal**, several signals share it - this cuts cost.
- Also when a channel would otherwise sit idle, as with bursty terminal traffic.
- The multiplexer combines n streams into one; the demultiplexer separates them again.

A32. Distinguish between the two basic multiplexing techniques. [Model]

Point	FDM	TDM
Resource shared	Frequency (a slice of bandwidth)	Time (a repeating slot)
Nature	Analog multiplexing	Digital multiplexing

Point	FDM	TDM
Guard band / gap	Guard bands between channels	Guard times or synchronisation bits
Simultaneous?	Yes - all channels at once	No - each channel takes its turn
Example	AM and FM broadcast, cable TV	Telephone PCM hierarchy

A33. What is a guard band? State its importance and its value for AM and FM. [Model]

- A guard band is an **unused slice of frequency** between two adjacent channels; it prevents the sidebands of one channel from bleeding into the next when filters are imperfect.
- **AM**: each station is allotted **10 kHz** and stations are spaced 10 kHz apart, so there is effectively no guard band.
- **FM**: each station is allotted **200 kHz** but the FCC has assigned only 50 of the 100 possible channels in the 88-108 MHz band, so each station gets a **200 kHz guard band**.

A34. Five channels, each with a 100 kHz bandwidth, are multiplexed with a guard band of 10 kHz. Find the minimum link bandwidth. [PYQ 2024-25; Notes]

1. Five channels need **four** guard bands (n-1, not n).
2. B = 5 x 100 + 4 x 10 = 500 + 40.
3. **B = 540 kHz.**

A35. Five channels, each with a 50 kHz bandwidth, are multiplexed with a guard band of 5 kHz. Find the minimum link bandwidth. [PYQ 2022]

1. B = 5 x 50 + 4 x 5 = 250 + 20.
2. **B = 270 kHz.** The trap is using 5 guard bands; only n-1 are needed.

A36. Three 4 kHz voice channels are to be combined into a 12 kHz link running from 20 to 32 kHz, with no guard bands. Show the configuration. [Notes]

1. Channel 1: modulate with a 20 kHz carrier, so it occupies **20-24 kHz**.
2. Channel 2: carrier 24 kHz, occupies **24-28 kHz**.
3. Channel 3: carrier 28 kHz, occupies **28-32 kHz**.
4. Each carrier sits at the lower edge of its band; the demultiplexer reverses the shift.

A37. Differentiate synchronous and asynchronous (statistical) TDM. [PYQ 2022; PYQ 2023; Model]

Point	Synchronous TDM	Statistical TDM
Slot allocation	Fixed slot per stream	Dynamic, on demand
Bandwidth use	Wasteful when a stream is idle	Efficient
Latency	Predictable	Varies with load
Complexity	Simple	Needs addressing and buffers
Suitable for	Continuous streams such as voice	Bursty computer data

A38. What is wavelength-division multiplexing? [Notes]

- An **analog multiplexing technique for optical fiber**: several light signals of different wavelengths (colours) share one fiber.
- A prism or diffraction grating combines and separates the wavelengths.
- It is FDM at optical frequencies, and it is what makes terabit backbones possible.

A39. Draw a suitable diagram to display packet transmission in a packet-switched network. [PYQ 2023]

- The message is **divided into packets** of fixed or variable size, each carrying a destination address.
- Each packet is handed to a router, which forwards it on its own - packets of one message may take **different routes**.
- They may arrive out of order, with different delays, and may be dropped if resources run out.
- Diagram: source on the left, three routers in the middle, destination on the right, with packets numbered 1, 2, 3 taking three different paths.

A40. What do you mean by a switched network? Write the requirements of a switching network. [Model]

- A switched network is a set of **switches connected by links**, where a switch creates a **temporary connection** between two or more devices for the duration of a conversation.
- Requirements: a way to **identify** the destination, a mechanism to **route or forward** the data to it, and a way to **release** the resources afterwards.
- Three kinds: circuit switching (physical layer), datagram packet switching (network layer), virtual circuit (data link layer).

A41. What is a virtual circuit? State the role of the VCI. [PYQ 2024-25]

- A virtual circuit is a **predefined path** set up between sender and receiver before any data moves; all packets follow that path and arrive in order.
- The **virtual-circuit identifier** is a small number carried in each frame that has **switch scope only** - it identifies the circuit on one link.
- A switch looks up its table by VCI instead of examining the full destination address, which makes forwarding fast.
- The table entry has four columns: input port, input VCI, output port, output VCI.

A42. Name the three phases of a virtual-circuit network. [PYQ 2024-25]

1. **Setup** - the source requests a connection using the **global addresses**; every switch creates a table entry.
2. **Data transfer** - frames carry the **VCI**, and each switch rewrites it for the next link.
3. **Teardown** - the source sends a teardown request, the destination confirms, and all switches delete the entry.

Section B | 7-mark questions

Write the heading points first, then expand each one. The bullet structure below is the marking scheme: one bullet is worth roughly half a mark to a mark.

B1. What is a transmission medium? Explain guided and unguided media with examples, and mention the advantages of optical fiber. [Model; PYQ 2022; PYQ 2023]

Definition. A transmission medium is anything that can carry information from a source to a destination - free space, a metallic cable or a fiber-optic cable.

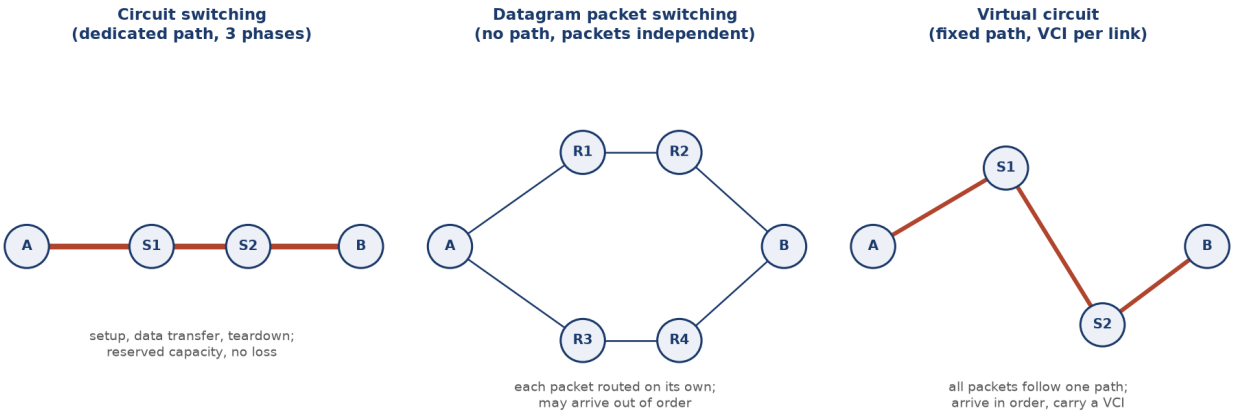
Medium	Construction	Bandwidth and distance	Application
UTP / STP	Twisted copper pairs, shielded or unshielded	Low to moderate, up to a few hundred metres	Telephone loop, Ethernet, LAN cabling
Coaxial	Core, dielectric, braided shield, jacket	Higher than twisted pair, longer runs	Cable TV, broadband Internet
Fiber-optic	Glass or plastic core with cladding, light source	Tbps, tens of kilometres between repeaters	Backbones, undersea cables

- **Guided media** provide a physical conduit: twisted pair, coaxial, fiber. The signal stays inside the cable, so they are secure and free of external interference (fiber completely).
- **Unguided media** carry electromagnetic waves without a conductor: **radio waves** (omni-directional, penetrate walls, multicast - radio and TV), **microwaves** (directional, line of sight, cannot penetrate walls - cellular, satellite, WLAN), **infrared** (short range, closed area, line of sight).
- **Fiber advantages**: Tbps bandwidth, low attenuation, immunity to EMI and crosstalk, no spark hazard, light and thin, very hard to tap.

Memory hook - guided = **T-C-F** (Twisted, Coaxial, Fiber); unguided = **R-M-I** (Radio, Microwave, Infrared), going from wide coverage to short range.

B2. Explain switched networks: circuit switching, datagram packet switching and virtual circuits, with diagrams. [PYQ 2022; PYQ 2023; PYQ 2024-25; Model]

Definition. A switched network is a collection of switches connected by links; a switch creates a temporary connection between two devices. The three kinds differ in *when* the path is decided.



1. **Circuit switching** - physical layer. The path is reserved before transmission and held for the whole conversation; the link is divided into channels by FDM or TDM; data flow continuously and are not packetised. Three phases: setup, data transfer, teardown. No delay at intermediate switches after setup, but capacity is wasted when idle.
2. **Datagram packet switching** - network layer. The message is divided into packets, each with the full destination address; no resource is reserved and packets are handled first-come-first-served. Packets may take different routes, arrive out of order or be dropped; the upper layer reorders or retransmits. The switches are routers.
3. **Virtual circuit** - a cross between the two. A path is set up before transfer, so packets arrive in order, but capacity is not reserved. Each packet carries a VCI, and switches use a table instead of the full address. Same three phases as circuit switching.

Point	Circuit switching	Packet switching
Path	Dedicated, reserved	None (datagram) or fixed (virtual circuit)
Layer	Physical	Network (datagram), data link (virtual circuit)
Delay	Setup delay, then none	Queueing delay per packet
Bandwidth use	Wasted while silent	Shared, efficient
Order and loss	Guaranteed order, no loss	May reorder or drop

B3. Explain the three phases of a virtual-circuit network and the addressing it uses. [PYQ 2024-25]

- **Two kinds of address**: a **global address**, unique in the whole network, used only during setup; and a **local VCI**, which has switch scope and is used for every frame after that.
- **Setup phase** - the source sends a setup frame with the global source and destination addresses; as it travels, each switch creates a table entry (input port, input VCI, output port, output VCI) and picks a free VCI for the next link.
- **Data-transfer phase** - every frame carries a VCI; a switch reads the VCI on the incoming port, finds the row, rewrites the VCI and sends it out of the recorded port.
- **Teardown phase** - the source sends a teardown request, the destination replies with a confirmation, and every switch deletes the entry, freeing the VCI.

Why the VCI is small: it only has to be unique on one link, not in the whole network, so 12 or 16 bits replace a full 32-bit address in every frame - a big saving in header overhead.

B4. Explain the key features of FDM and TDM, and distinguish synchronous from asynchronous TDM.

[PYQ 2022; PYQ 2023; Model]

- **FDM** - analog multiplexing: the bandwidth of the link is divided into bands, one per signal, separated by guard bands; all signals travel at once. Used in AM and FM broadcast, cable TV and the first-generation telephone carrier systems. Each channel is shifted to its own carrier frequency.
- **TDM** - digital multiplexing: time is divided into frames, and each frame into slots, one per channel; all channels use the whole bandwidth but take turns. Used in the digital telephone hierarchy (DS-0 to DS-4).
- **Synchronous TDM** assigns a fixed slot to each stream whether or not it has data, so slots are wasted; timing is simple and delay is predictable.
- **Asynchronous (statistical) TDM** allocates slots on demand, so no slot is wasted, but each slot must carry an address and the receiver needs buffers; delay varies with load.

Point	Synchronous TDM	Statistical TDM
Slot	Pre-assigned, fixed	Assigned only when data are present
Efficiency	Low for bursty traffic	High
Overhead	Almost none	Address per slot
Delay	Constant	Variable

B5. In a TDM system with 32 channels, each channel carrying 8 bits per sample at a sampling rate of 8 kHz, calculate the total data rate. [PYQ 2024-25]

1. Each channel needs 8 bits x 8000 samples per second = **64 kbps**.
2. The frame carries one sample from each of the 32 channels: 32 x 8 = **256 bits per frame**.
3. Frames are sent 8000 times per second.
4. Total rate = 32 x 8 x 8000 = **2,048,000 bps = 2.048 Mbps**.
5. This is the E-1 / DS-1 line rate, and it explains why one telephone channel is 64 kbps.

B6. What is multiplexing? Explain FDM, WDM and TDM with their applications. [Model]

Definition. Multiplexing allows the simultaneous transmission of several signals over one link. The multiplexer combines n streams into one (many-to-one) and the demultiplexer separates them (one-to-many).

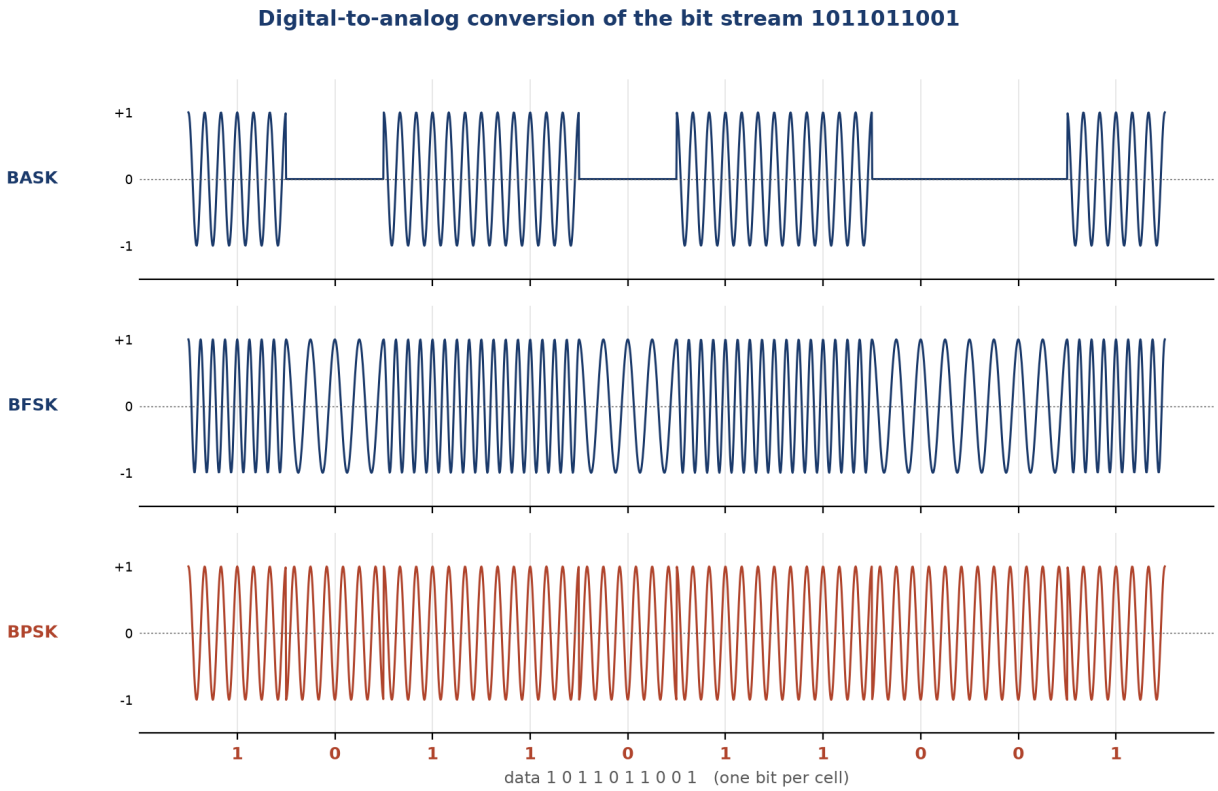
- **FDM** - shares the bandwidth; each signal is modulated onto a different carrier and guard bands separate the channels. Applications: AM and FM radio, cable TV, first-generation telephone systems.
- **WDM** - FDM for optical fiber; different wavelengths of light share one fiber, combined and split by a prism. Application: fiber backbones, where a single fiber carries many 10 Gbps channels.
- **TDM** - shares time; the link carries one channel at a time in repeating slots. Applications: digital telephone hierarchy, SONET/SDH, and any digital trunk.
- **Inverse multiplexing** - the opposite: one high-rate stream is spread over several low-rate links.

Exam one-liner: FDM divides the *spectrum*, TDM divides the *clock*, WDM divides the *colour*.

B7. Explain digital-to-analog conversion. Sketch BASK, BFSK and BPSK for the data stream 1011011001.

[PYQ 2024-25; PYQ 2022; Model]

Principle. Digital-to-analog conversion changes one characteristic of a carrier - C(t) = A cos(2 pi fc t + phi) - according to the digital data. Bit rate N and baud rate S are related by S = N x 1/r, and the minimum bandwidth is B = (1 + d)S.



1. **BASK** - amplitude switched: 1 sends the carrier, 0 sends zero. Generated by multiplying unipolar NRZ with the carrier. Weakest against noise.
2. **BFSK** - frequency switched: 1 sends f1 = fc + delta-f, 0 sends f2 = fc - delta-f. Generated with NRZ and a VCO. Constant amplitude, widest bandwidth.
3. **BPSK** - phase switched: 1 sends phase 0, 0 sends phase 180. Generated by multiplying polar NRZ with the carrier. Best noise immunity, same bandwidth as ASK.

How to score the sketch: draw the carrier axis, mark the bit cells, then show the carrier full for 1 and zero for 0 (BASK), dense and sparse cycles (BFSK), and upright versus inverted cycles (BPSK). Label fc, f1, f2 and the bit boundaries.

B8. Explain binary phase shift keying in detail. [PYQ 2024-25 backlog]

- **Definition**: BPSK varies the phase of the carrier between 0 and 180 degrees according to the bit; the amplitude and frequency stay constant.
- **Generation**: multiply the polar NRZ signal with the carrier - a +V level leaves the carrier unchanged, a -V level inverts it.
- **Constellation**: two points on the I-axis at +1 and -1, separated by 180 degrees; the minimum distance is 2, the largest possible for two points on a unit circle, which is why BPSK is the most noise-immune of the binary schemes.
- **Bandwidth**: B = (1 + d)S, identical to ASK, with r = 1 so one bit per symbol.
- **Detection**: coherent - the receiver must recover the carrier phase and multiply by a locally generated carrier, then integrate over the bit interval.
- **Use**: deep-space links, satellite, and the base of QPSK, which sends 2 bits per symbol.

Compare in one line: ASK varies A, FSK varies f, BPSK varies phi - and noise attacks A hardest, so BPSK wins on error rate while FSK wins on simplicity.

B9. Explain analog-to-analog conversion: AM, FM and PM, with bandwidths and applications. [Model]

Why modulate. Analog information such as voice needs no conversion to be sent baseband, but a bandpass medium or a shared medium requires the signal to be shifted onto a carrier.

1. **AM** - the amplitude of the carrier follows the modulating signal, so the modulating signal becomes the **envelope** of the carrier. Bandwidth = 2B. Simple and cheap, but the most susceptible to noise. Application: AM broadcast, 535-1605 kHz, 10 kHz per station.

2. **FM** - the frequency of the carrier follows the modulating signal; amplitude stays constant. Bandwidth = 2(1 + beta)B, so it needs more spectrum but rejects amplitude noise. Application: FM broadcast, 88-108 MHz, 200 kHz per station.
3. **PM** - the phase of the carrier follows the modulating signal. The phase change appears as a frequency change, but the instantaneous frequency shift is proportional to the derivative of the modulating signal. Bandwidth is the same form as FM. Application: used inside digital schemes and in some satellite links.

Point	AM	FM	PM
Parameter varied	Amplitude	Frequency	Phase
Bandwidth	2B	2(1 + beta)B	2(1 + beta)B
Noise immunity	Poor	Good	Good
Hardware	Simple	Complex	Complex

B10. Explain the constellation diagram and multilevel modulation with QAM. [PYQ 2024-25 backlog]

- A constellation diagram plots each symbol as a point in the I-Q plane: the **distance from the origin** is the amplitude and the **angle** is the phase.
- With L points each symbol carries **log2 L bits**, so the bit rate rises without widening the band - 16-QAM carries 4 bits per symbol.
- **QAM** combines ASK and PSK: L amplitude levels times M phases give L x M points. Its bandwidth is the same as ASK, which is why modems, digital cable TV and Wi-Fi use it.
- The **minimum distance between points** decides noise immunity: as more points are packed in, the distance falls, so a higher SNR is needed for the same error rate.
- PSK constellations put all points on a circle (constant envelope, cheaper amplifiers); QAM puts them on a grid (better distance for the same power).

Exam line: more points = more bits per symbol = more capacity, but the same bandwidth and a stricter SNR requirement. That trade-off is the whole of digital modulation.